HOUSSEM EDDINE MAGRA

MSc Student in Data Science & Intelligent Systems | Software Engineer

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PROFILE

Master's student in Data Science and Intelligent Systems at Université Constantine 2, with a Bachelor degree in Software Engineering and hands-on experience in machine learning, data analysis, and software development. Interested in explainable machine learning and data-driven methods for cardiovascular risk monitoring. Combines an applied AI background with experience building web, mobile, and data-oriented applications.

EDUCATION

Master 2 in Data Science and Intelligent Systems | In progress, expected July 2027
Université Constantine 2 – Abdelhamid Mehri, Constantine, Algeria

Master 1 in Data Science | Completed
Université Constantine 2 – Abdelhamid Mehri, Constantine, Algeria

Bachelor's Degree (Licence) in Software Engineering
Université Constantine 2 – Abdelhamid Mehri, Constantine, Algeria

RESEARCH EXPERIENCE

Clinical Risk Modeling and Explainability: Mortality Prediction in Heart Failure Patients | Master 1 project

- Studied mortality prediction using a clinical dataset of 299 heart-failure patients and 11 routinely collected clinical variables.
- Implemented and compared machine-learning approaches, including Logistic Regression, Random Forest, LightGBM, and CatBoost, as well as a soft-voting ensemble.
- Explored feature engineering, probability calibration using Platt scaling, and model interpretation with SHAP.
- Developed an initial applied research workflow for clinical risk modeling; further validation on larger and independent cohorts remains necessary.

Sentiment Analysis on Tourism Platforms | Python, scikit-learn, Keras

- Worked on review-data exploration, cleaning, preprocessing, and feature engineering for tourism-platform sentiment analysis.
- Contributed to logistic-regression and deep-learning model experiments, evaluation, and result visualisation.

SELECTED TECHNICAL PROJECTS

ChargeOps Sys and AI – Intelligent Wireless EV Charging Monitoring System | Python, PyTorch, FastAPI, SQLite, Docker

- Developed an integrated prototype for monitoring wireless power-transfer systems for electric vehicles, combining system monitoring, simulated telemetry, and AI-based diagnostics.
- Implemented neural-network-based fault diagnosis and Remaining Useful Life (RUL) estimation to support predictive maintenance.
- Contributed to the system architecture, integrating an edge simulator, Python API backend, data storage, and a real-time monitoring dashboard.

RIF Connect – International Conference Mobile Application | Flutter, Mobile Application Development

- Contributed to the development of RIF Connect, the official mobile application associated with the RIF International Conference (Recherche en Informatique au Féminin), held in Constantine, Algeria, on 8–9 December 2025.
- The project aimed to enhance the conference experience through a centralized mobile platform for communication and access to event information.

PROFESSIONAL EXPERIENCE

Radio Engineering Intern / Web Developer – Ooredoo Algérie | Constantine, 2026

- Completed an internship within the radio engineering team, gaining exposure to mobile-network architecture and radio access technologies from 2G to 5G.
- Worked with the radio-planning team after the internship on web-development tasks and technical problem-solving.
- Explored site planning, sectorisation, handover optimisation, radio parameters, network KPIs, and coordination with operational and transmission teams.

TECHNICAL SKILLS

Programming: Python, Java, Dart, Kotlin, JavaScript, SQL
Machine Learning & Data: PyTorch, scikit-learn, Keras, data preprocessing, exploratory data analysis, model evaluation, SHAP
Development: FastAPI, Django REST, Flutter, React, REST APIs, PostgreSQL, SQLite
Tools & Platforms: Git/GitHub, Docker, Jupyter, Firebase, Supabase, GitHub Actions, Render

CERTIFICATIONS & RECOGNITION

- Python Certification – Kaggle
- English Language Certification – B2, ULC School, Constantine
- Certificate of Appreciation – Mozilla Club, for teaching Database Fundamentals to university students
- Certificate of Completion and Appreciation – RIF Conference, for contribution to the RIF Connect application
- Internship Certificate – Ooredoo Algérie, Mobile Networks
- Certificate of achievement Building with Artificial Intelligence - Saylor University
- Certificate of participation Hackathon Forsatic - Algeria Telecom 2025

LANGUAGES

Arabic (native) | French (fluent) | English (B2 certified; fluent)